Technology Stack

Date	03 July 2026
Team ID	SWTID-2026-4350
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*HTML, CSS, JavaScript*	*Used to create a simple, responsive web interface for entering applicant details, displaying prediction results, and viewing dashboard information.*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*Python, Flask, Flask-SQLAlchemy*	*Flask handles the web application, request processing, routing, and integration with the Machine Learning prediction module. Flask-SQLAlchemy manages database operations using ORM.*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*SQL Database (via SQLAlchemy)*	*Stores user information, applicant details, credit history, ML model information, and loan prediction results using relational database tables.*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*Flask Development Server / Waitress (Production)*	*The application runs using the Flask development server during development and can use the Waitress WSGI server in a production environment, as described in the project.*
5	**Version Control & CI/CD**	***Git and GitHub*** *(if used for your project)*	*Used to maintain source code versions, collaborate with team members, and manage project updates. If you did not use GitHub, you can write* ***"Not Used"*** *instead.*

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*CreditPredictor Module(src/prediction.py)*	*Integrates the Machine Learning prediction service to generate loan approval predictions and assign risk categories based on applicant data.*